UNREAL ENGINE PLUGIN

Screen Waypoint Markers

A marker that follows something in the world and sticks to the edge of the screen when it goes out of frame — including when it is behind you.

Unreal Engine 5.6, 5.7, 5.8 · **Setup time** about two minutes · **Blueprint** everything in this guide

1 The two-minute version

> **INSTALLING**
>
> In the Epic Games Launcher, go to Library → Fab Library, find **ACGA Screen Waypoint Markers**, choose **Install to Engine** and pick your engine version. It arrives already compiled, so a Blueprint-only project works as it is.
>
> In your project, open Edit → Plugins, search for *ACGA Waypoint*, tick it and restart the editor. To see its assets, open the Content Browser's Settings and tick **Show Engine Content** and **Show Plugin Content**.

> **PRESS PLAY BEFORE YOU READ THE REST**
>
> The plugin ships a finished, playable level. Open `ACGAWaypoint → Demo → Maps → DemoMap_ACGA_Waypoint` and press Play. There is already a character on it, with the controls set up and the level told to use it, so nothing in your project needs configuring first — and nothing in your project is changed.
>
> **Turn around before you do anything else.** There is a waypoint directly behind where you start, and its marker was already stuck to the edge of your screen on the very first frame. That is the thing this plugin exists to get right, and it is the thing most others get wrong.
>
> WASD and the mouse move and look. **K** switches every waypoint in the level off and back on. There are seven of them, spread around a walled district; one of them, off to your left behind a wall, changes colour while the wall is in the way.

If you only read one page, read this one.

Drag `BP_ACGA_Waypoint` out of the plugin's Blueprints folder into your level, type a name on it, and press Play. That is a working waypoint.

ACGAWaypoint ▸ Blueprints ▸ BP_ACGA_Waypoint

To put one on something that already exists — a chest, an NPC, a door:

1. **Select the actor**
   and choose Add Component →
   ACGA Screen Waypoint
   .

2. **Move the component up**
   in the viewport, so the marker floats above the thing rather than inside it.

3. **Type a Label**
   and pick an
   Icon Id
   from the dropdown.

It already draws something. You do not have to build a widget first, and there is nothing to switch on.

ONE ARRANGEMENT TO AVOID

Do not make the waypoint component the **root** of your actor — keep a scene component or a mesh as the root and attach the waypoint under it. Section 6 explains why. The shipped `BP_ACGA_Waypoint` is already built that way, so copying it keeps you safe.

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2 Placing and configuring

Where the marker sits

The component is a normal scene component. **Move it where you want the marker to appear.** There is no separate "offset" setting to hunt for — if the marker should float two metres above a crate, move the component two metres up.

The settings you will actually touch

SETTING	WHAT IT DOES
Label	The text drawn on the marker. Leave it empty for no text.
Icon Id	Which icon, from a dropdown of what is in the icon set.
Start Active	Untick for a marker that appears later — when a quest is accepted, when a place is discovered. Turn it on with *Set Waypoint Active*.
Clamp To Screen	Whether the marker sticks to the edge of the screen when the thing is out of frame, or just disappears off the side. Normally on.
Clamp Shape	Rectangle follows the edges of the screen; ellipse follows an oval. Rectangle is what most HUDs want.
Clamp Margin	How far in from the edge the pinned marker sits.
Visibility Mode	**Always**; **When On Screen** only while the thing is actually in frame; **When Clamped** only while it is off frame, which makes a pure off-screen arrow.
Max Visible Distance	Hide it past this distance, in centimetres. 0 means never hide.
Fade Start Distance	Start fading out here, reaching invisible at the distance above.
Priority	When too many markers want the screen at once, higher wins. Ties go to whichever is nearer.
Check Occlusion	Notice when something is between the camera and the thing — a wall, a building.
Hide When Occluded	With the above on: disappear behind walls instead of just dimming.

> **TOO MANY MARKERS ON SCREEN**
>
> Project Settings → Plugins → ACGA Waypoint → `Max On Screen` caps how many are drawn at once. Use `Priority` on the important ones to decide who survives the cap.

3 Your own icons

Icons live in one asset, and adding yours takes about a minute.

The plugin ships with a set at:

ACGAWaypoint ▸ Icons ▸ DA_ACGA_WaypointIcons_Default

It contains `Default`, `Question`, `Alert` and `Dot`.

1. **Duplicate it**
   into your own Content folder, so a plugin update cannot overwrite your work.

2. **Add a row**
   , give it a name and a texture. The name is what you pick from the dropdown, so make it readable — `Vendor`, not `Icon_07`.

3. **Point the project at your copy:**
   Edit → Project Settings → Plugins → ACGA Waypoint → `Default Icon Set`.

Your names now appear in the `Icon Id` dropdown on every waypoint.

> **A NOTE ABOUT THE DROPDOWN**
>
> The list is built from the **project's** icon set. If you give one component its own icon set it still works perfectly at runtime, but its names will not be the ones offered in the list.

> **SAME NAMES AS THE MAP**
>
> The ACGA Minimap plugin has an icon set built the same way with the same starting names, so you can say `Question` once and mean it in both. They are separate assets so that either plugin can be removed without breaking the other.

4 Changing how it looks

Three levels, in increasing effort. Most projects stop at the first.

Level one — edit a style

A **style** asset holds everything that is not the icon: the arrow, the colours for each state, the fonts, and how the distance is written. Duplicate the shipped one and point Project Settings at your copy.

ACGAWaypoint ▸ Styles ▸ DA_ACGA_WaypointStyle_Default

GROUP	WHAT IS IN IT
Icon	A fallback icon, and whether to hide the icon while the marker is stuck to the edge.
Arrow	The arrow art, when it shows, and how far out from the marker it orbits. **Arrow Clears Marker** pushes it past the text automatically, which is what you want when your labels vary in length.
Colour	Three tints — normal, stuck to the edge, and hidden behind something — plus how faint a distant marker is allowed to get.
Text	Whether to draw the label and the distance, their fonts and colours, and how the distance is written. **Distance Format** takes `{0}` for the number: write `{0} ft` and set **Distance Scale** to work in feet.

> **DRAW YOUR ARROW POINTING UP**
>
> The marker rotates your arrow art to point at the target. It expects the art to point **straight up** to begin with. Get that right and there is nothing to correct — no offset setting, no "invert rotation" tickbox.

Level two — several styles

Make more than one style asset and set `Style` on individual components. Quest markers in gold, points of interest in grey.

Level three — your own widget

Make a UMG widget, set `Widget Class` on the component to it, and it replaces the shipped marker entirely. Make it a child of **ACGA Waypoint Widget** and it comes with the useful values already available in the graph: whether the marker is stuck to the edge, which way to point, how far away the thing is, and whether it is behind a wall.

5 Showing and hiding markers

Untick `Start Active` to author a marker that begins hidden, then call **Set Waypoint Active** when it should appear.

> **PREFER THIS OVER THE VISIBLE TICKBOX**
>
> Unticking the component's own *Visible* also hides it, but then the plugin still thinks the waypoint is on and *Is Waypoint Active* disagrees with the screen. `Start Active` keeps the two in step.

To hide every marker at once — a cutscene, a photo mode — use **Set All Waypoints Active**.

6 The behind-you problem

Worth two minutes of reading, because it is the thing this plugin exists to solve and it explains one rule you have to follow.

When something is behind the camera, Unreal's own screen-marker system stops drawing it. Not "draws it in the wrong place" — stops entirely. That is why so many off-screen indicators simply vanish the moment you turn around, which is exactly when you need them most.

This plugin works around it, and the marker slides smoothly around the edge of the screen as you turn. The workaround needs one thing from you:

> **THE RULE**
>
> The waypoint component must not be the **root component** of its actor. Give the actor a scene component or a mesh as its root, and attach the waypoint underneath. In the Components panel, the waypoint should be indented under something else, not at the very top.

If you copy `BP_ACGA_Waypoint`, or add the component to an actor that already has a mesh, this is automatic. The plugin writes a warning to the log if it ever finds otherwise.

7 Every setting, in one place

Project Settings → Plugins → ACGA Waypoint

SETTING	WHAT IT DOES
Update Hz	How often distances and occlusion are re-checked. The marker itself still follows the camera every frame, so this does not make it laggy.
Max On Screen	Most markers drawn at once. 0 means no limit.
Occlusion Traces Per Pass	How much work the "is it behind a wall" check may do each pass. Raise it if that state feels slow to update.
Default Style	The look every waypoint uses unless given its own.
Default Icon Set	The icons every waypoint can choose from.
Default Marker Widget Class	Your own widget, applied to every waypoint still using the shipped one.
Debug Draw	Prints what each waypoint is doing, over the game. See section 8.

8 When something does not look right

Open the console with the tilde key and type:

```
ACGA.Waypoint.DebugDraw 1
```

You get a line per waypoint listing its state, which way it is pointing, how far away it is, **which widget was made for it** and **which icon it resolved to**. Those last two answer most questions on their own.

WHAT YOU SEE	WHAT TO CHECK
Nothing at all	The readout says `widget NONE` : the component has no Widget Class, or the component is disabled.
A marker with no icon	The readout says `icon NONE` : the Icon Id is not a name in the set, or that row in the set has no texture.
It vanishes when the thing is behind me	The waypoint component is the actor's root. See section 6.
The arrow sits on top of the text	Turn on `Arrow Clears Marker` in the style, or raise `Arrow Distance` .
Markers flicker on and off	`Max On Screen` is lower than the number competing. Raise it, or use Priority to decide who wins.
Labels overlap at the screen edge	Markers pinned to the same edge are not spread apart in this version. Lower `Max On Screen` and raise `Priority` on the ones that matter, or set `Visibility Mode` to *When On Screen* on markers that need no off-screen indicator.
A marker never appears	`Start Active` is unticked and nothing ever called *Set Waypoint Active*.

Removing the plugin

Delete the components and untick the plugin. It does not replace any engine class and does not need anything in your GameMode or Player Controller, so a project that removes it still opens.